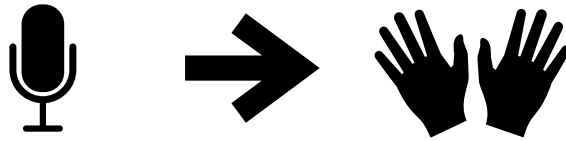


Disignum: Speech to Sign AI



Problem Statement

There are around 70 million deaf people around the world and 430 million people experience hearing loss. The deaf and hard-of-hearing community face significant challenges in daily life because most people communicate verbally rather than through sign language. This creates a huge gap in conversations, making it hard for deaf people to chat with friends, teachers, doctors, or at work without help.

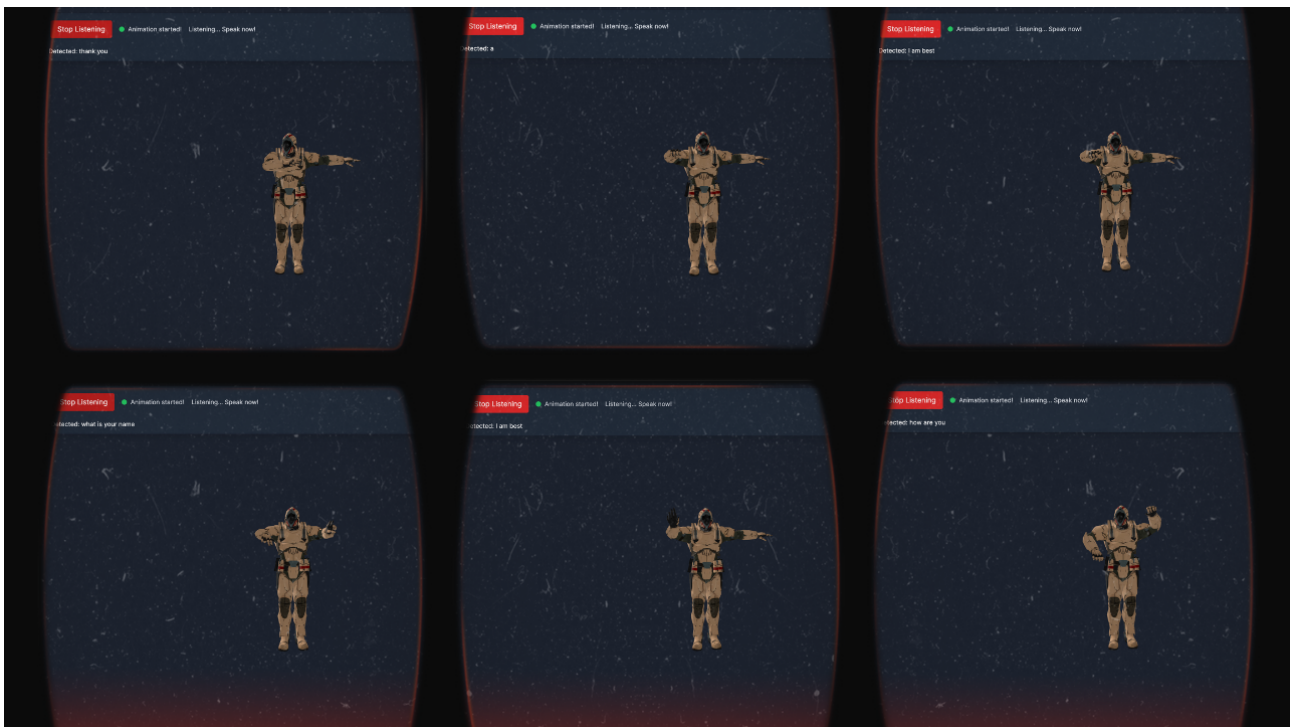
Some Major Problems that We see are:-

- The global cost of unaddressed hearing loss is estimated at **US \$1.3 trillion** annually, largely due to lost productivity.
- With an estimated 18 million deaf or hard-of-hearing individuals in India, there is roughly only one interpreter for every 50,000 to 70,000 deaf persons.
- 85% of these 339 interpreters are located in just five cities (Delhi, Mumbai, Hyderabad, Bengaluru, and Chennai), leaving over 600 districts in India with zero professional coverage.

Old tools like human interpreters or expensive video translators are **not practical**. They are costly, slow, and not always available, especially for poor families. These solutions don't work in real time and can't handle back-and-forth talks well. This leaves deaf people dependent and stops them from joining online classes, events, or jobs fully. That is where our project Disignum Shines. It is a low-cost, real-time speech-to-sign language converter bridging the communication gap between the deaf and common people. That is where our project Disignum Shines. It is a low-cost, real-time speech-to-sign language converter bridging the communication gap between the deaf and common people.

The Solution: Disignum App

To address this challenge, We present Disignum, an AI-powered application that converts speech into American Sign Language in real time using 3D models. Disignum is designed to bridge the communication gap for the deaf and hard-of-hearing community, it help users to fully participate in conversations, lectures, and events without delay. Our project can help Hard of Hearing people to live more peacefully and enjoyably in the society. It can function as a daily translator, allowing the deaf and hard-of-hearing community to interact with teachers, doctors, peers, and society without delay or dependency. It can translate complete video/audio lectures to ASL for Deaf people.

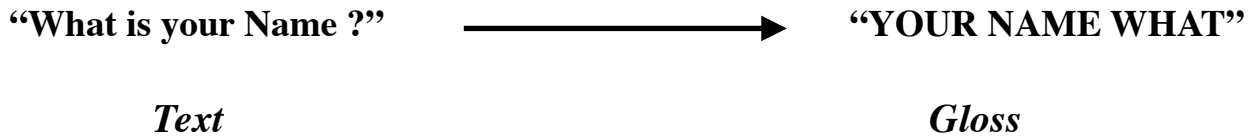


App prototype

Working of Disignum

We designed a new and affordable way to translate speech into sign language, reducing the overall cost and making it accessible for everyone — especially those from poor and underprivileged backgrounds.

Our system starts by using **speech recognition** to convert spoken words into text. Then, a **pretrained text-to-gloss model** transforms that text into sign glosses (the written form of sign language). After that, these glosses are translated into a **sequence of movements**, which are then applied to a **3D avatar** that performs the signs in real time.



The most unique part of our system lies in the **gloss-to-movement translation**. For this, we built and **trained our own AI model from scratch** using a Transformer architecture. We trained it on our own dataset which we created by extracting poses from videos and saving to JSON file. These videos are from the open-source How2Sign dataset, which contains sign language videos along with their corresponding sentences. This helped us create our **own lightweight dataset**, reducing the data size and training cost — while keeping the same quality and accuracy.

Our trained AI model outputs pose landmarks for each gloss in MediaPipe format. These landmarks are then converted into joint angles using **inverse and forward kinematics**, which are finally applied to a 3D avatar to perform the correct signs.

Through this process, we have created a tool that acts as both a **speech translator** and a **sign language interpreter** — a true bridge between the hearing and the deaf communities.

It empowers deaf and hard-of-hearing individuals to **communicate freely** without delay, dependency, or discrimination. This innovation brings inclusion, dignity, and connection into the hands of those who have long been left unheard.

Stages	Technology	Description
1. Speech to text	Speech Recognition	The system first captures spoken words and accurately transcribes them into written text.
2. Text to Gloss	Pre-trained Transformer Model	The transcribed text is then transformed into sign glosses— the written representation of sign language.
3. Gloss to Pose	Custom Trained AI	This is the core of our innovation. The sign glosses are translated into a precise sequence of body, hand, and facial movements.
4. Pose to Animation	3D Rendering	The movement data is applied to a realistic 3D avatar, which performs the signs in real time.

Unique Value

Disignum is a mobile-integrated, two-way communication platform that provides real-time sign language translation with zero training barriers.

True Two-Way Dialogue: While competitors like **Hand Talk** or **SignAll** may focus on one-way translation, **Disignum** is the only platform that supports **Two-way Communication**.

This allows for a natural, back-and-forth conversation between a deaf individual and a hearing person, rather than just a one-sided broadcast.

Scalable for Business: Because it offers an API for Businesses, companies can integrate this two-way, low-cost communication directly into their own apps or customer service portals, making the world more accessible at scale.

Team and Vision

While governments are creating laws to empower their citizens, there is a significant shortage of human interpreters, and we find ourselves at the centre of this golden opportunity.

By harnessing the power of AI, our team of student researchers, advised by professors and our non-profit and for-profit structure, is working towards making it ready for the market.

Our team includes me and my brother as co-founders(I am a passionate AI researcher, and my brother is a Finance and tech learner), Tanishq as our third co-founder, and 8-9 students + 1 professor under the Research department that is operated as a non-profit.

This structure helps us reduce cost, maximise the impact, and collect our proprietary dataset for our AI model. Disignum is not just a business; it is also a community service, and our approach towards it helps us attract more people, professors, and industry leaders.